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 Studierichting: Informatica
 Oefeningenreeks 2E nr. 4.16

$$\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{a}{x^2} \right)^x$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{ax}{x^2} + \frac{a}{x^2} \right)^x$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{ax + a}{x^2} \right)^x$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{a(x+1)}{x^2} \right)^x$$

$$\text{Neem } \frac{1}{k(x)} := \frac{a(x+1)}{x^2} \Rightarrow k(x) = \frac{x^2}{a(x+1)}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{k(x)} \right)^x$$

$$\text{Omdat } x = \frac{a(x+1)}{x^2} \cdot k(x)$$

$$\lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{k(x)} \right)^{k(x)} \right]^{\frac{a(x+1)}{x^2} \cdot x}$$

$$= e^a$$

References